

API Integration Guide with Energy Scheduling Platform

API INTEGRATION GUIDE

ACCENTURE SOLUTIONS PRIVATE LIMITED

Document Revision History

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S. No	Date	Author	Reviewer	Approver	Version	Remarks
1	29-Apr-24	Accenture Team	Rajprakash Yadav	Pralay Saha Pushpak Ghosh Dastidar	1.0	Initial Version

Abbreviation List

Abbreviation	Expansion
API	Application Programming Interface
AS	Ancillary Service
CSV	Comma-separated values
DAM	Day Ahead Market
GNA	Green Network Access
TGNA	Temporary Green Network Access
GDAM	Green Day Ahead Market
HTTP	Hyper Text Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IP	Internet Protocol
JSON	JavaScript Object Notation
MW	Mega Watt
NA	Not Available
HPX	Hindustan Power Exchange
MBAS	Metering, Billing, and Accounting System
POSOCO	Power System Operation Corporation Limited
PX	Power Exchange
HPDAM	High Price Day Ahead Market
REST	Representational State Transfer
RTM	Real Time Market
SSL	Secure Sockets Layer
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
IEX	Indian Energy Exchange

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1. API Format

APIs for integrating will be published as stateless REST API over HTTPS.

2. API Security

API security is applied at the network and application level. The application security can be explained as follows:

HTTPS: The communication between the API shall take place a secure https layer SSL protocol. This way communication will happen through a secure encrypted connection.

IP Whitelisting: Whitelisting of IP will be done at gateway level. This will ensure access control through network.

Authorization Basic Auth

Username

Password

3. Energy Scheduling API List

3.1 Get Schedule Data

Get schedule data

API Description	API details	Example	Remarks
URI	{{baseHREF}}reports/1.0/WebAccess API/GetUtilityExternalSharedData?api key={{apiKey}}		
HTTP Method	POST		
Content Type	Application/json		
Accept	Application/xml		
Header/API Key	X-API-Key		The API Key can be sent in one of the following ways: i. As an HTTP Header named X-API-Key ii. As a URL query parameter named

API Description	API details	Example	Remarks
			apikey
Request Attributes JSON	<pre>{ "Date": "", "SchdRevNo": -1, "UserName": "", "UtilAcronymList": [] }</pre>	<pre>{ "Date": "29-04-2024", "SchdRevNo": -1, "UserName": "test", "UtilAcronymList": ["KUDGI"] }</pre>	Date should be in dd-mm-yyyy format. Username should match with provided API credentials
Response Attributes JSON		<pre>{ "ResponseBody": { "Date": "29-04-2024", "FullSchdRevisionNo": 0, "ScheduleRemarks": "test_sch_1", "ApiRemarks": null, "SchedulePublishedTime": "2024-04-29T16:40:41", "GroupWiseDataList": [{ "Acronym": "KUDGI", "FullschdList": [{ "EnergyScheduleTypeName": "OA_GNA", "EnergyScheduleSubTypeName": "NA", "ASTypeName": "NA", "DataIntegrationTimingTypeName": "NA", "PXTransactionTypeName": "NA", "PXExchangeTypeName": "NA", "SellerAcronym": "KUDGI", "SellerUnitAcronym": "", "SellerParentStateUtilAcronym": "", "SellerParentDiscomUtilAcronym": null, "IsUnitWiseSellerFlag": 0, "BuyerAcronym": "APTRANSCO", "BuyerUnitAcronym": "", "BuyerParentStateUtilAcronym": "ANDHRA_PRADESH_STATE", "BuyerParentDiscomUtilAcronym": null, "IsUnitWiseBuyerFlag": 0, "ApprovalNo": "GNA/SR/2024/" }] }] } }</pre>	

[illegible]

[illegible]

[illegible]

API Description	API details	Example	Remarks
		0, 0, 0, 0,0,0,0,0,0,0,0,0,0], "OffbarLPSCRegAmount": [0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0 ,0 ,0 ,0 ,0,0,0,0,0,0,0,0,0], "OnbarRegOASoldAmount" :[0, 0, 0, 0, 0,0,0,0,0,0,0,0,0], "OnbarUnRegOASoldAmo unt":[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0, 0, 0, 0, 0, 0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0] ,"OnbarFinalEntAmount":["0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0 ,0 ,0 ,0 ,0,0,0,0,0,0,0,0,0], "OffbarFinalEntAmount": [0, 0, 0, 0, 0,0,0,0,0,0,0,0,0]}}, "GasEntJsonData": null, "Nuc learEntJsonData": null, "HydroEntJsonData": nul l}, "EntitlementRevisionNo": 5}], "RequisitionLis	

API Description	API details	Example	Remarks
		"NuclearReqJsonData":null,"HydroReqJsonDa ta":null},"RequisitionRevisionNo":0}], "Declara tionList":[{"UtilAcronym":"KUDGI","UnitAcron ym":"","IsUnitWiseData":0,"RevisionNo":0,"Cr eatedOn":"2024-04- 26T17:34:46","GeneratorTypeName":"THERM AL","GeneratorSubTypeName":"11","Declarati onData":{"ThermalDCJsonData":{"SellerInpOn barAmount":[2240,2240,2240,2240,2240,22 40,2240,2240,2240,2240,2240,2240,2240, 2240,2240,2240,2240,2240,2240,2240,2240 ,2240,2240,2240,2240,2240,2240,2240,224 0,2240,2240,2240,2240,2240,2240,2240,22 40,2240,2240,2240,2240,2240,2240,2240,2 240,2240,2240,2240,2240,2240,2240,2240, 2240,2240,2240,2240,2240,2240,2240,2240 ,2240,2240,2240,2240,2240,2240,2240,224 0,2240,2240,2240,2240,2240,2240,2240,22 40,2240,2240,2240,2240,2240,2240,2240,2 240,2240,2240,2240,2240], "SellerInpOffbar Amount":[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0, 0, 0, 0, 0, 0, 0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0] }, "SellerInpTotalAm ount":[2240,2240,2240,2240,2240,2240,224 0,2240,2240,2240,2240,2240,2240,2240,22 40,2240,2240,2240,2240,2240,2240,2240,2 240,2240,2240,2240,2240,2240,2240,2240, 2240,2240,2240,2240,2240,2240,2240,2240	

[illegible]

[illegible]

[illegible]

API Description	API details	Example	Remarks
		0,1500,1500,1500,1500,1500,1500,150 0,1500,1500,1500,1500,1500,1500,150 0,1500,1500,1500,1500,1500,1500,150 0,1500,1500,1500,1500,1500,1500,150 0,1500,1500,1500,1500,1500,1000,2250,100 0,1000,1000,1000,1000,1000,1000,1000,10 00,1000,1000,2250,2250,2250,2250,2250,2 250,2250,2250,2250,2250], "MaxInstallCapaci ty":0,"OverLoadingPercentage":5,"Normative DCPossibleValuesList":null,"DefaultTechMinP ct":55,"BypassNormativeDCCheck":0},"GasD CJsonData":null,"NuclearDCJsonData":null,"H ydroDCJsonData":null,"RenewableDC.JsonDat a":null}}]]}}, "ResponseStatus":{"Code":"WBES- 200","Message":null,"DetailsList":[]}	
Error Attributes	JSON HTTP 200: Success HTTP 400: Bad Request HTTP 401: Unauthorized HTTP 404: Not Found	{ "ResponseBody": null, "ResponseStatus": { "Code": "WBES-400", "Message": "API Access Validation Failed - ", "DetailsList": [>User credentials created not matched with passed username: passed user - and credential user - noar-integration",	

API Description	API details	Example	Remarks
		<pre> "User credentials not found with username: . Please contact Power Grid Admin."] } } </pre>	

3.1.1 REST Client Samples

3.1.1.1 cURL

```

curl --location {{BaseHref}}reports/1.0/WebAccessAPI/GetUtilityExternalSharedData?apikey={{apiKey}} \
--header 'Content-Type: application/json' \
--header 'Authorization: Basic bm9hci1pbnRlZ3JhdGlvbjpOb2FyQDIwMjQ=' \
--header 'Cookie: Path=/' \
--data '{
  "Date":"26-04-2024",
  "SchdRevNo": -1,
  "UserName":"noar-integration",
  "UtilAcronymList": [],
  "UtilRegionIdList":[1,2,3,4,5]
}'

```

4.7.1.2 C# (.Net)

```

var options = new RestClientOptions("")
{
    MaxTimeout = -1,
};
var client = new RestClient(options);
var request = new RestRequest("<BASE_HREF>/POSOCO/reports/1.0/WebAccessAPI/GetUtilityExternalSharedData?apikey==X",
Method.Post);
request.AddHeader("Content-Type", "application/json");
request.AddHeader("Authorization", "Basic c2NoZWZy5pbnRlZ3JhdGlvbWpTY2hlZHVsaW5nQDAyMjAyNA==");
var body = @"{
" + "\n" +
@"    ""Date"":""26-04-2024"",
" + "\n" +
@"    ""SchdRevNo"": -1,
" + "\n" +
@"    ""UserName"":""noar-integration"",
" + "\n" +
@"    ""UtilAcronymList"": [],
" + "\n" +
@"    ""UtilRegionIdList"":[1,2,3,4,5]
" + "\n" +
@"}";
request.AddStringBody(body, DataFormat.Json);
RestResponse response = await client.ExecuteAsync(request);
Console.WriteLine(response.Content);

```

3.1.1.3 Java

```

OkHttpClient client = new OkHttpClient().newBuilder()
    .build();
MediaType mediaType = MediaType.parse("application/json");

```

```

RequestBody body = RequestBody.create(mediaType, "{\r\n  \"Date\": \"26-04-2024\", \r\n  \"SchdRevNo\": -1, \r\n  \"UserName\": \"noar-integration\", \r\n  \"UtilAcronymList\": [], \r\n  \"UtilRegionIdList\": [1,2,3,4,5] \r\n}");
Request request = new Request.Builder() .url("<BASE_HREF>/reports/1.0/WebAccessAPI/GetUtilityExternalSharedData?apikey=X")
    .method("POST", body)
    .addHeader("Authorization", "Basic c2NoZWV1bGluZy5pbmRlZ3JhdGlvbW5nQDAyMjAyNA==")
    .method("POST", body)
    .addHeader("Content-Type", "application/json")
    .build();
Response response = client.newCall(request).execute();

```

3.1.1.4 NodeJS

```

const axios = require('axios');
let data = JSON.stringify({
  "Date": "26-04-2024",
  "SchdRevNo": -1,
  "UserName": "noar-integration",
  "UtilAcronymList": [],
  "UtilRegionIdList": [
    1,
    2,
    3,
    4,
    5
  ]
});
let config = {
  method: 'post',
  maxBodyLength: Infinity,
  url: '<BASE_HREF>/reports/1.0/WebAccessAPI/GetUtilityExternalSharedData?apikey=X',
  headers: {
    'Authorization': 'Basic c2NoZWV1bGluZy5pbmRlZ3JhdGlvbW5nQDAyMjAyNA=='
  }
}

```

```
    },  
    data: data  
  };  
  
  axios.request(config)  
    .then((response) => {  
      console.log(JSON.stringify(response.data));  
    })  
    .catch((error) => {  
      console.log(error);  
    });
```

3.1.1.5 Python

```
import http.client  
  
conn = http.client.HTTPSConnection("<BASE_HREF>")  
json.dumps({  
  "Date": "26-04-2024",  
  "SchdRevNo": -1,  
  "UserName": "noar-integration",  
  "UtilAcronymList": [],  
  "UtilRegionIdList": [  
    1,  
    2,  
    3,  
    4,  
    5  
  ]  
})
```

```
headers = {  
    'Authorization': 'Basic c2NoZWV1bGluZy5pbnRIZ3JhdGlvbjpTY2hlZHVsaW5nQDAyMjAyNA=='  
}  
conn.request("POST", reports/1.0/WebAccessAPI/GetUtilityExternalSharedData?apikey=X", payload, headers)  
  
res = conn.getresponse()  
data = res.read()  
print(data.decode("utf-8"))
```